Лабораторна робота №6

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Група ІПЗ-21

Варіант 7

Код консольної програми

```
using Lab6_Semeniuk_07;

class Program
{
   static void Main()
   {
       const string divider =
"------------------------------------------------------------";

       Console.WriteLine(divider);
       Console.WriteLine("Fullname: Vlad Semeniuk");
       Console.WriteLine("Group: IPZ-21");
       Console.WriteLine("Variant: 7");
       Console.WriteLine(divider);


       Cylinder firstObject = new Cylinder("Сталь", 2.5, 8);
       Cylinder secondObject = new Cylinder("Алюміній", 1.2,
4.5);


       Console.WriteLine("Initial Objects:");

       Console.WriteLine();
       firstObject.PrintInfo();
       Console.WriteLine();
       secondObject.PrintInfo();
       Console.WriteLine(divider);
```

```
        firstObject.Scale(2.5);
        secondObject.Scale(2.5);



        Console.WriteLine("After 2.5 scale:\n");

        firstObject.PrintInfo();
        Console.WriteLine($"Surface Area
{firstObject.CalculateSurfaceArea():F2}");
        Console.WriteLine();
        secondObject.PrintInfo();
        Console.WriteLine($"Surface Area
{secondObject.CalculateSurfaceArea():F2}");
        Console.WriteLine(divider);

        Console.WriteLine("Scale only first object:");
        firstObject.Scale(5);

        Console.WriteLine();
        firstObject.PrintInfo();
        Console.WriteLine();
        secondObject.PrintInfo();
        Console.WriteLine("The second object didn't change
because we create two instances of one class (Cylinder) with
new keyword which create a separate instance of a class.");
        Console.WriteLine(divider);

        Cylinder testObject = firstObject;

        Console.WriteLine("Scale first object together with
testObject:");
        firstObject.Scale(5);
```

```
        Console.WriteLine();
        firstObject.PrintInfo();
        Console.WriteLine();
        testObject.PrintInfo();

        Console.WriteLine();
        Console.WriteLine("The state is the same in both
objects");
        Console.WriteLine("Because we assign firstObject with =
which gives the same link to the same instance of the object
and we didn't use new keyword so a new instance was not
created.");
        Console.WriteLine(divider);

        Console.WriteLine("Individual test: A");

        Cylinder firstNewObject = new
Cylinder("FirstTestObject");
        Cylinder secondNewObject = new
Cylinder("SecondTestObject", 2.2, 3.3);

        firstNewObject.PrintInfo();
        Console.WriteLine();
        secondNewObject.PrintInfo();
        Console.WriteLine();

        Console.WriteLine("Our C# compiler knows about two
constructors and it can decides which one to use");
        Console.WriteLine("If we put only material parameter
the compiler will choose the right one");
        Console.WriteLine("Initialization happens it the chosen
constructor");
```

```
    }
}
```

```
namespace Lab6_Semeniuk_07;

public class Cylinder
{
    public string Material;
    public double Radius;
    public double Height;

    public static int Counter = 0;

    public Cylinder(string material, double radius, double
height)
    {
        this.Material = material;
        this.Radius = radius;
        this.Height = height;
    }

    public Cylinder(string material)
        : this(material, 0, 0)
    {
        Console.WriteLine("The light constructor");
    }


    public double CalculateSurfaceArea()
    {
        return 2 * Math.PI * this.Radius * (this.Radius +
this.Height);
    }

    public void Scale(double factor)
    {
        if (factor > 0)
        {
            this.Radius *= factor;
            this.Height *= factor;
            return;
```

```
        }

        Console.WriteLine("The scale factor isn't valid.");

    }



    public void PrintInfo()
    {
        Console.WriteLine($"Material: {this.Material}");
        Console.WriteLine($"Radius: {this.Radius}");
        Console.WriteLine($"Height: {this.Height}");

    }


}
```

ШІ використовувався тільки для ресьорчу.